AGENTVERSE 2026 · ROBOTICS

Rig Robot

Two quadrupeds and **Argon**, the rig's control centre: rust repair and gas-leak response. They know when to call a human.



THE PROBLEM

Two chronic jobs that put people in the hazard zone

Corrosion

Pipework rusts constantly in salt spray. People fix it by hand from rope access, 30 m up, next to live hydrocarbons.

Small gas leaks

Bolted flanges work loose. A person with a gas detector walks up to the leak to find it.

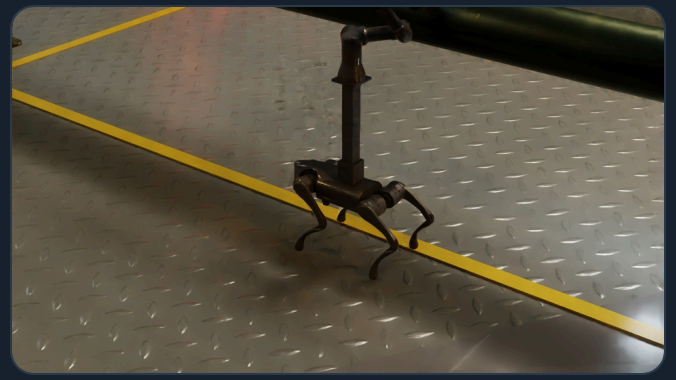
Dangerous · costly · never finished. User: asset-integrity and maintenance crews.

The robot must never **paint over rust** or **put a wrench on a cracked flange.**

WHAT WE BUILT

A two-robot fleet, coordinated by Argon

On a full offshore rig in Isaac Sim, rendered on an NVIDIA GB10. Both robots and every fixed gas detector report to Argon.



Unit 1 · Scrubber

Go2 + lift mast + UR3, brush pad, spray and paint nozzle, gas sniffer. Spray → soak → scrub (PPO) → verify → paint.



Unit 2 · Fixer

Go2 + lift mast + UR3 with socket tool, gas sensors. Lidar walk-in → sniff → inspect bolts → re-torque → verify → dock.

Argon · gemini-4-argon

- Fuses each fixed detector with the nearest robot's sniffer
- Sends the robot whose tools fit the job
- Holds hot work near gas
- Files tickets for a human; never approves its own repair

Fixed detectors sit on every flange: green · amber · red · grey for a faulty head. Safety rules act locally and at once; Gemini's replies are checked before anything moves.

DEMO · ONE TAKE · ≤ 60 S

▶ Play video

Unit 1 sprays and scrubs → a bolt loosens at F5 (scripted, not announced) → detector amber: spark rule holds Unit 1 → Argon sends Unit 2 → leak confirmed → re-torque → green → Argon resumes Unit 1 → paint

Also in the run: a stuck detector gets a ticket, not a repair · Argon's premature “resume” is refused while the leak is open

File: **TBD** (runs/fleet/fleet_video.mp4) · fallback: the fleet HUD replay of the live Argon run

ARGON · THE CONTROL CENTRE

Two sensors, one verdict

FIXED DETECTOR	ROBOT SNIFFER	VERDICT	ARGON
High	High	**LEAK_CONFIRMED**	Unit 2 fixes the flange
High	Clean, beside it for 3 s	**PIPE_SENSOR_FAULT**	Ticket to a human, no repair
Clean	High	**LEAK_UNSEEN_BY_FIXED**	Inspect, flag the detector
High	Too far away to tell	**UNVERIFIED**	Send Unit 2 to check
Clean	Clean	**OK**	Carry on

18/18
scenario evals correct

4/4
valid live calls in the demo run

4–9 s
per call, non-blocking

≈ $0.003
per call

Hazard first: ≥ 20 %LEL is always real gas. **Spark rule:** hot work stops within 6 m of an amber or red detector. Link cut, timeout or a bad reply → the same rules decide locally.

PERCEPTION-DRIVEN ACTION

Pixels become the work plan

RUST · MOBILENETV4-S + LORA, 1.1 MS ON CPU

Frame → 96 px patches → Heatmap → Ray–cylinder → (θ, z) grid on pipe → Spray · scrub · done check

0.977
F1, 881 real photos

13/13
held-out Isaac renders

LEAK · NO ML: TORQUE STRIPES + CRACK CHECK

Frame → Project known bolts → Read each red stripe → Thin dark shapes, mm length → View quality → Loose · crack · unsure

40/40
cracks found

0/40
false alarms

≈ 1°
bolt-angle error

SCRUB POLICY · PPO IN ISAAC LAB, 2,048 ENVS

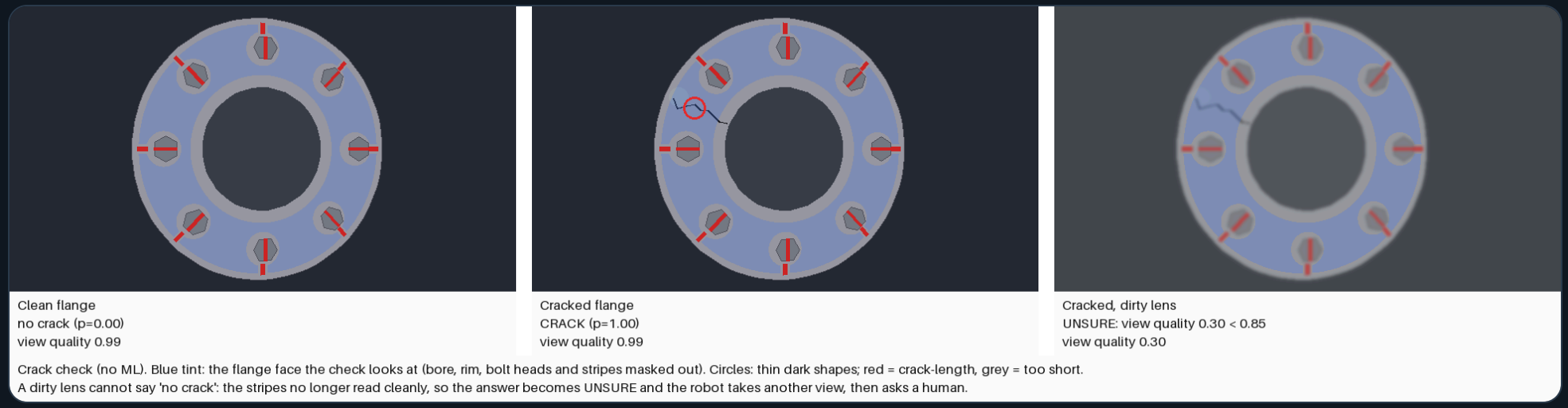
v1 cleaned everything by pressing too hard

RIG GEOMETRY, NEW RUST PATCH EACH EPISODE	V1	V2 (FORCE-LIMITED)
Success (< 5 % rust left)	100 %	97.6 %
Episodes over the 40 N limit	**76.8 %**	**1.0 %**
Mean / p95 episode-max force	**49.9 / 70.1 N**	**29.9 / 35.7 N**
Mean time to clean	5.2 s	6.8 s

Wear model: removal ~ force × speed, 4× faster after a 30 s soak. v2 adds a quadratic over-force penalty and a 38 N setpoint cap. The v2 demo render peaks at 31.9 N.

GAS-LEAK JOB · UNIT 2

Loose bolt, or crack? The camera decides



Loose bolts 1 and 5

Re-torque, stripe aligned, gas clears → DONE

Cracked flange

Back off 3 m, critical alert → ESCALATE

Dirty lens

“No crack” not trusted → 2 extra views → ESCALATE

Leak near hot work

Unit 1 holds, Unit 2 re-torques, Argon resumes Unit 1 → DONE

CRITICAL FAILURE MODES

The most severe hazard always wins

Paint over rust	< 5 % rust, ≥ 80 % visible · Gemini veto · 3 cycles → ESCALATE
Wrench on a crack	crack possible → no bolt touched · back off 3 m
Gas ≥ 20 %LEL · robot sensor fault	**GAS_RETREAT** (fail-safe)
Detector amber within 6 m of hot work	Spark rule: **HOLD** Unit 1 at once · Unit 2 fixes · resume
Gemini says “resume” with the leak open	Refused and logged (happened in the live run)
Link to Argon cut	Same rules decide locally
Force > 40 N	**BACK_OFF**: retract 5 cm, × 0.7
Person < 3 m · lens blocked	**PAUSE_PERSON · HOLD**



The HUD in mock replay: BACK_OFF at 65 N > 40 N, with the decision log and pipe map.

Check order before every step: gas sensor working → gas below limit → gas clear → person clear.

EDGE AUTONOMY · WHAT'S NEXT

The fast loop never leaves the robot

On the robot · CPU

- Rust detector: 1.1 ms
- Scrub policy: 0.012 ms*
- Leak check: ~90 ms / frame*
- Lidar navigation · safety rules
- GB10 TensorRT: **TBD**

Cloud · Argon

- gemini-4-argon fleet decisions
- Non-blocking · 4–9 s per call
- Gemini rust second opinion
- Offline → edge rules make the safe call

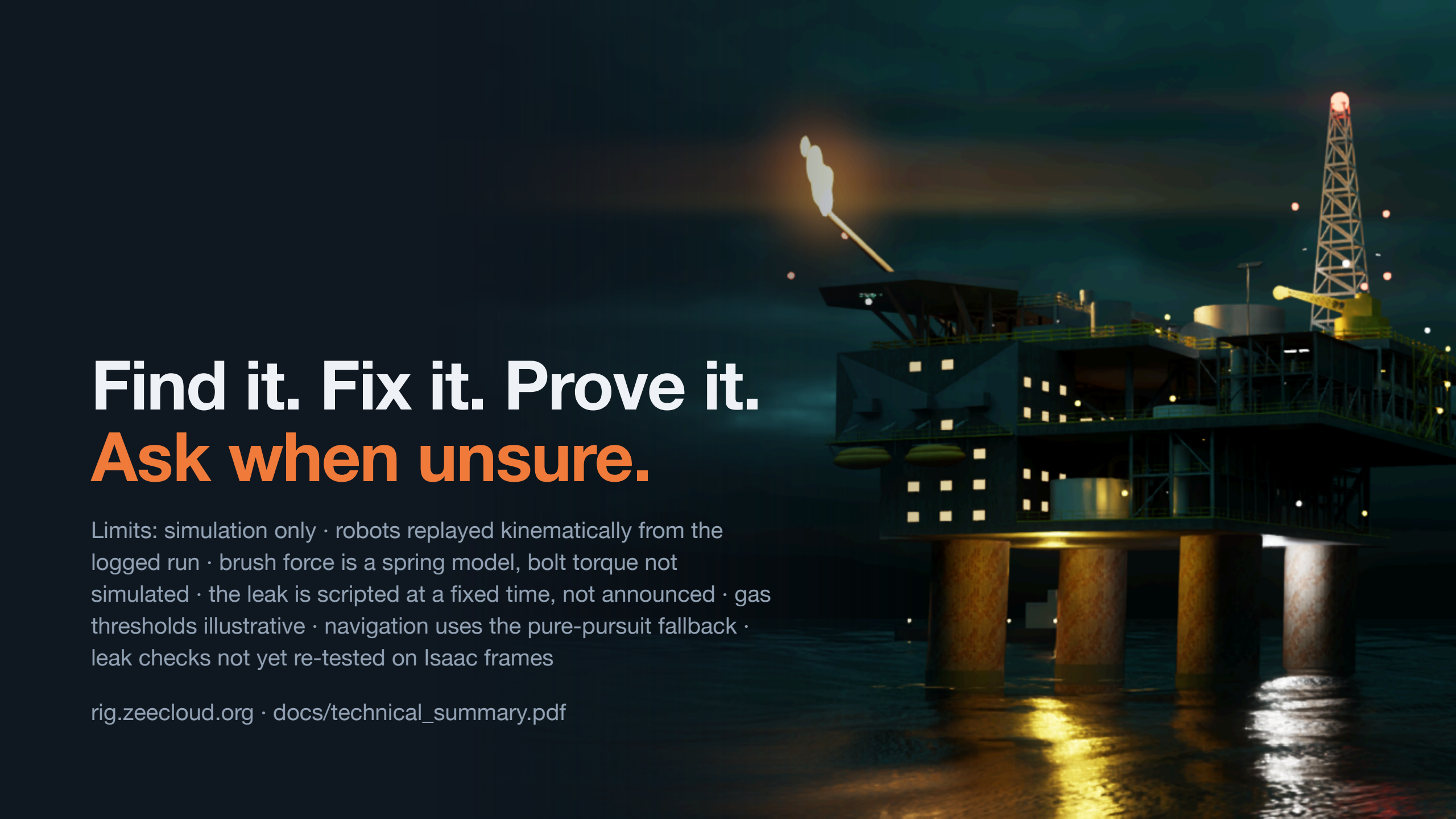
Training only

- Modal A10G · L40S
- GB10: PPO + RTX render + stream

Next

Trained PPO lidar navigation on both units (the demo uses the pure-pursuit fallback) · 3D gas plume with orifice leak rate · storm adaptation · more units and jobs under Argon

*measured on a laptop CPU, batch 1.



Find it. Fix it. Prove it.
Ask when unsure.

Limits: simulation only · robots replayed kinematically from the logged run · brush force is a spring model, bolt torque not simulated · the leak is scripted at a fixed time, not announced · gas thresholds illustrative · navigation uses the pure-pursuit fallback · leak checks not yet re-tested on Isaac frames

rig.zeecloud.org · docs/technical_summary.pdf